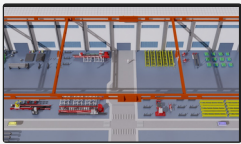


1 Digital-twin line



Isaac Sim water-pipe line

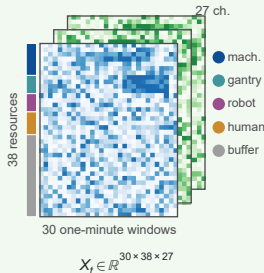


Human Machine Material Logistics

one disturbance axis per episode

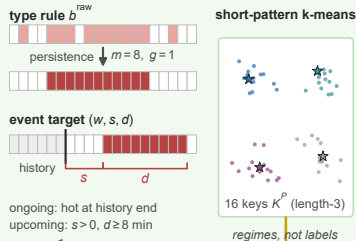
2 Multi-entity state

C1



3 Process labels + clustering

C2

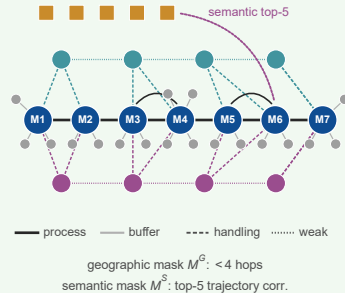


training targets

K^P

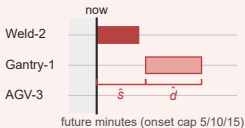
4 Heterogeneous graph

C1



X_t, G, M^G, M^S

6 Bottleneck report



process cause $\hat{c}$



remaining order time $\hat{t}$

12 min left

5 PHAST-DP

process-informed heterogeneous-graph Transformer

Multi-task heads

- occurrence $p^{\text{cont}} / p^{\text{onset}}$
station event
- start π^s , duration $\hat{d}$
when and how long
- prefix $\hat{p}$, occupancy $\hat{O}$
dense future grid
- cause $\hat{c}$, remain $\hat{t}$
graph-level pooling
- validation-controlled decoding

Encoder block $\times 5$ ($d=80$)

Concat + project

$[H_T \parallel H_G \parallel H_S]$

Add & Norm

+ feed-forward

Pattern attention

queue patch $P_{t,n} \times$ activity keys $K^P \rightarrow C^P$

Geographic attention (2 heads)

mask M^G ; keys + C^P (delayed propagation)

Temporal attention (4 heads)

each resource across 30 windows

Semantic attention (2 heads)

mask M^S ; similar but non-adjacent resources

Grouped embedding

Queue	4
Stall	6
Logistics	4
Shortage	1
Context	7
Type	5

$$+ E_{\text{time}} + E_{\text{lap}} + E_{\text{node}} \rightarrow H^{(0)}$$